THE MILKY WAY

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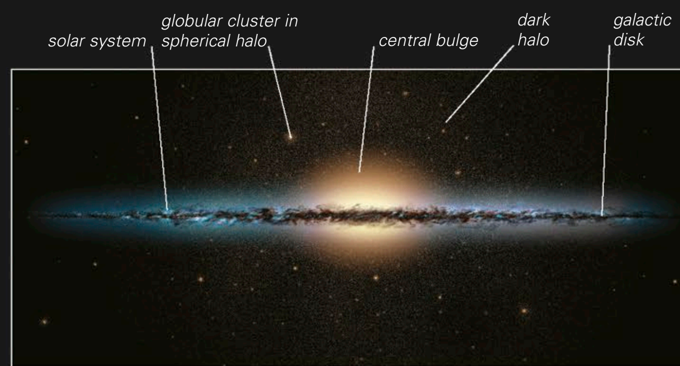
THE SUN IS ONE STAR of around 200 billion that make up the Milky Way, a relatively large spiral galaxy (see p.302) that started to form around 13.5 billion years ago. From our position inside the Milky Way, it appears as a bright band of stars stretching across the night sky.

BAND OF STARS
As we look out along the disk of the Milky Way from our position within, we see a bright band of thousands of stars that has captured humankind's imagination throughout history.

THE GEOGRAPHY OF THE MILKY WAY

At the very center of the Milky Way lies a black hole with a mass of about 4 million solar masses. This core or nucleus of the galaxy is surrounded by a bulge of stars that grows denser closer to the center. This forms an ellipsoid of about 15,000 by 6,000 light-years, the longest dimension lying along the plane of the Milky Way. Lying in the plane is the disk containing most of the galaxy's stellar materials. Young stars etch out a spiral pattern, and it is thought that they radiate out from a bar. Surrounding the bulge and disk is a spherical halo in which lie some 200 globular clusters, and this in turn may be surrounded by a dark halo, the corona.



MILKY WAY GALAXY
The Milky Way has a diameter of about 180,000 light-years and a thickness of about 2,000 light-years. The Sun lies about 26,000 light-years from the center.